Anatomy of brain stem and associated lesions:

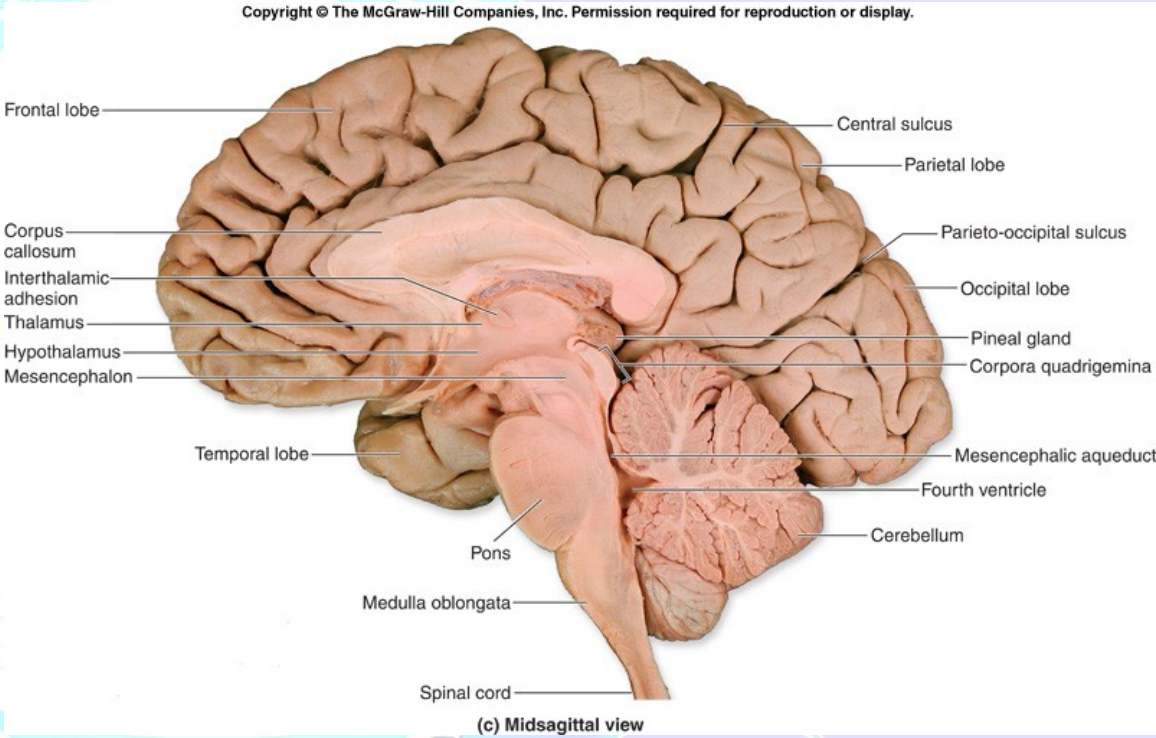
PONS

Learning objective

- At the end of the lecture the student should be able to know:
- External features of pons.
- Internal structure of pons.
- Lesions of pons.

Brain stem

- Underneath the limbic system is the brain stem.
- This structure is responsible for basic vital life functions such as breathing, heartbeat, and blood pressure.
- The brain stem is made of
  - Midbrain
  - Pons
  - Medulla.



Pons

- Forepart of the hind-brain
- Situated in front of the cerebellum
- Behind and below, the pons is continuous with the medulla oblongata, but is separated from it in front by a furrow in which the abducent, facial, and vestibulocochlear nerves appear.

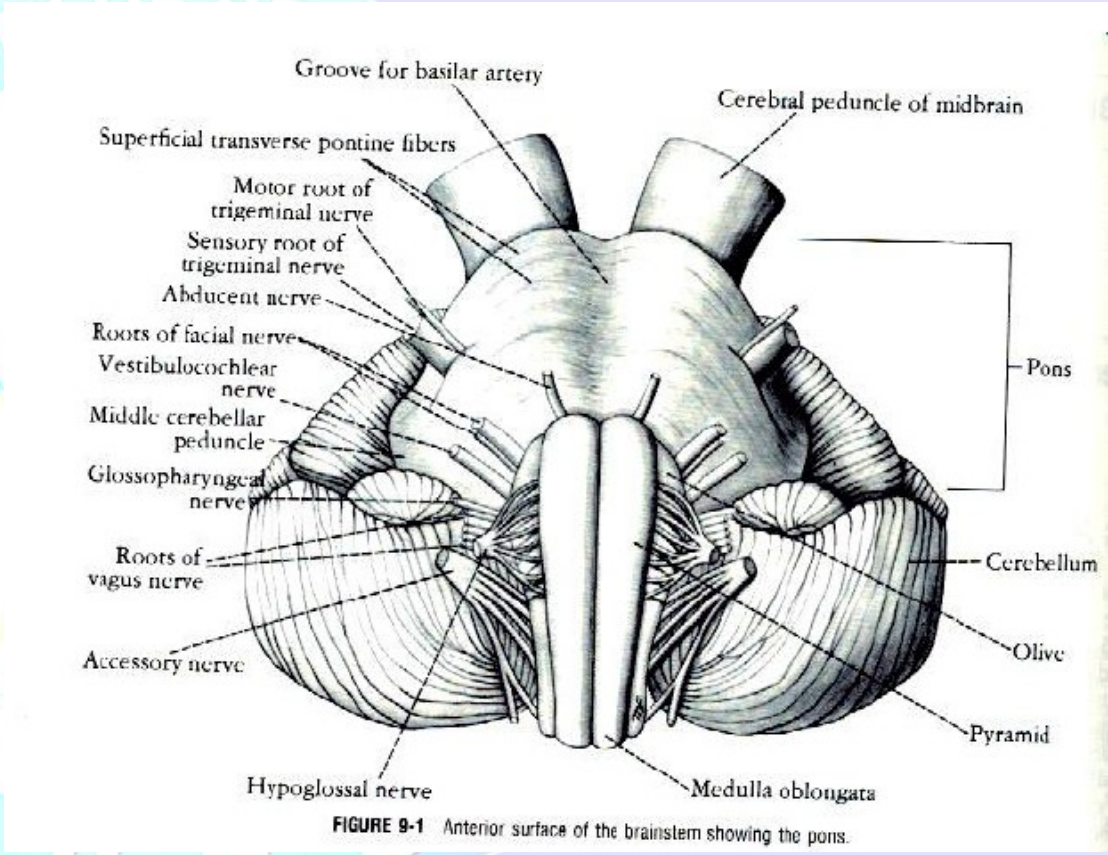


FIGURE 9-1 Anterior surface of the brainstem showing the pons.

Pons

- Associated with cranial nerves:
  - **Trigeminal (V) chewing**
  - **Abducens (VI) movement of**
  - **Facial (VII)- facial expressio**
  - **Vestibulcochlear (VIII) –bala**

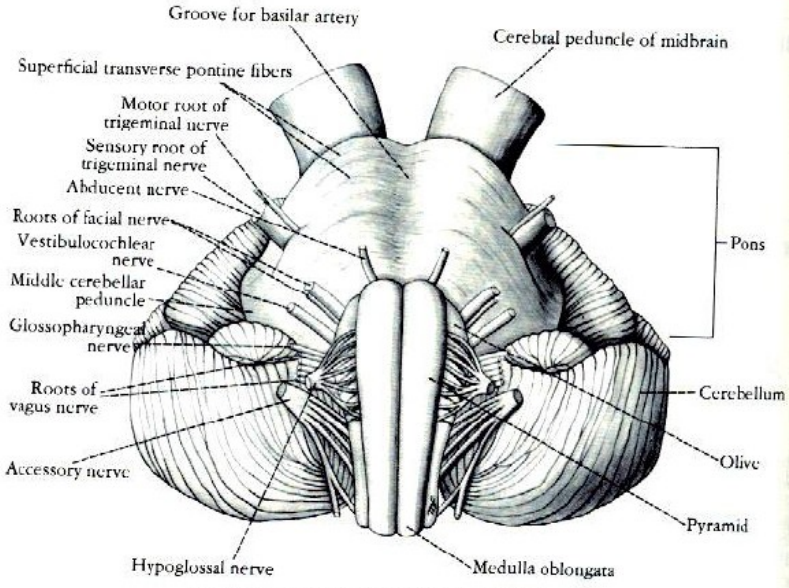


FIGURE 9-1 Anterior surface of the brainstem showing the pons.

Ventral or **Anterior surface** of **pons**

- Basilar sulcus
  - Midline sulcus occupied by basilar artery
- Transverse running fibres or pontocerebellar fibres which cerebellar peduncle.
- Trigeminal nerve emerge at

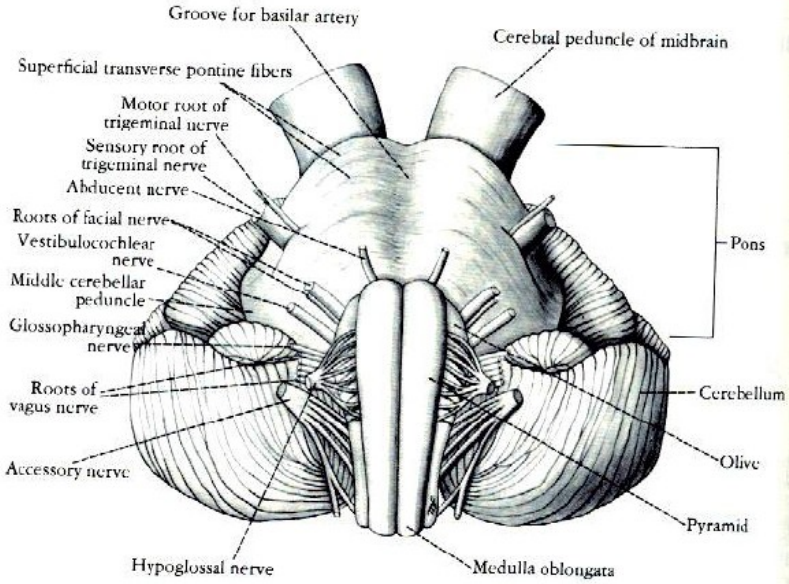


FIGURE 9-1 Anterior surface of the brainstem showing the pons.

Dorsal or **Posterior surface** of **Pons**

- Hidden by cerebellum
- Bounded laterally by superior cerebellar peduncle
- Median sulcus in the midline
- Facial colliculus
  - Paramedian elevation raised by underlying abducent nerve covered by winding fibres of facial nerve.
- Trochlear nerve

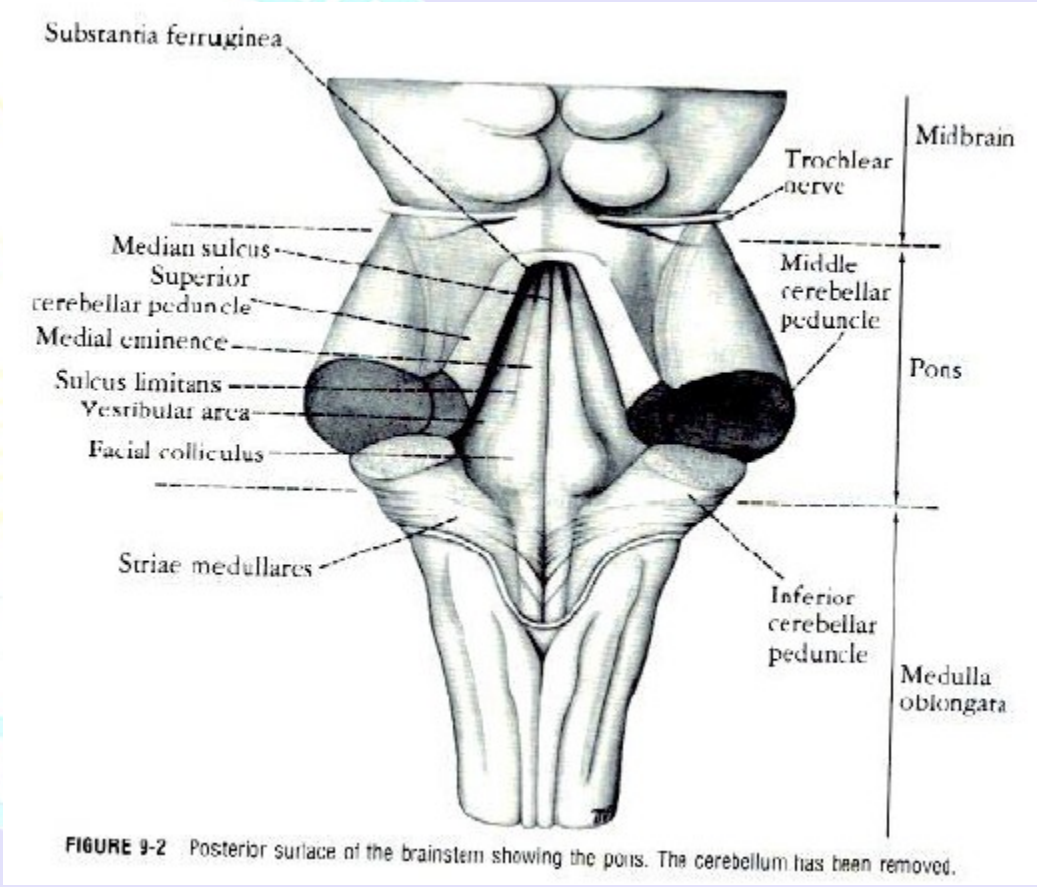
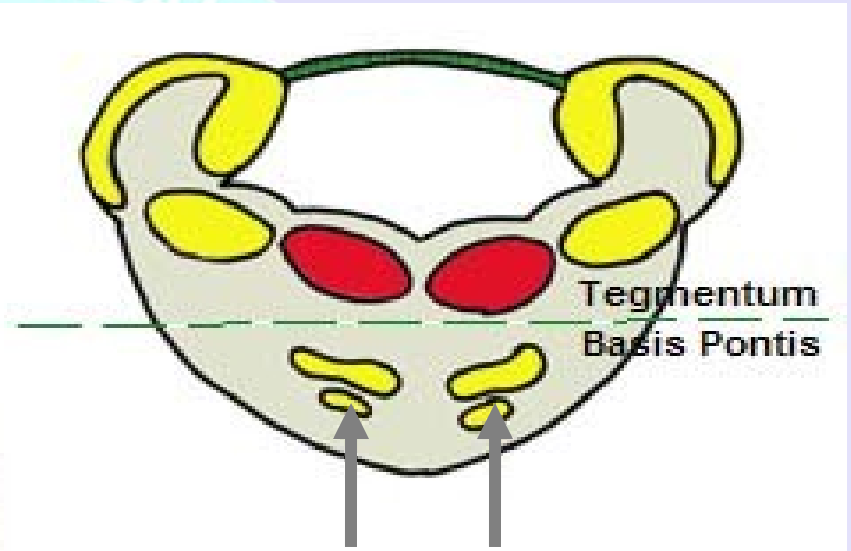


FIGURE 9-2 Posterior surface of the brainstem showing the pons. The cerebellum has been removed.

Internal structure of pons

- Basilar or ventral portion

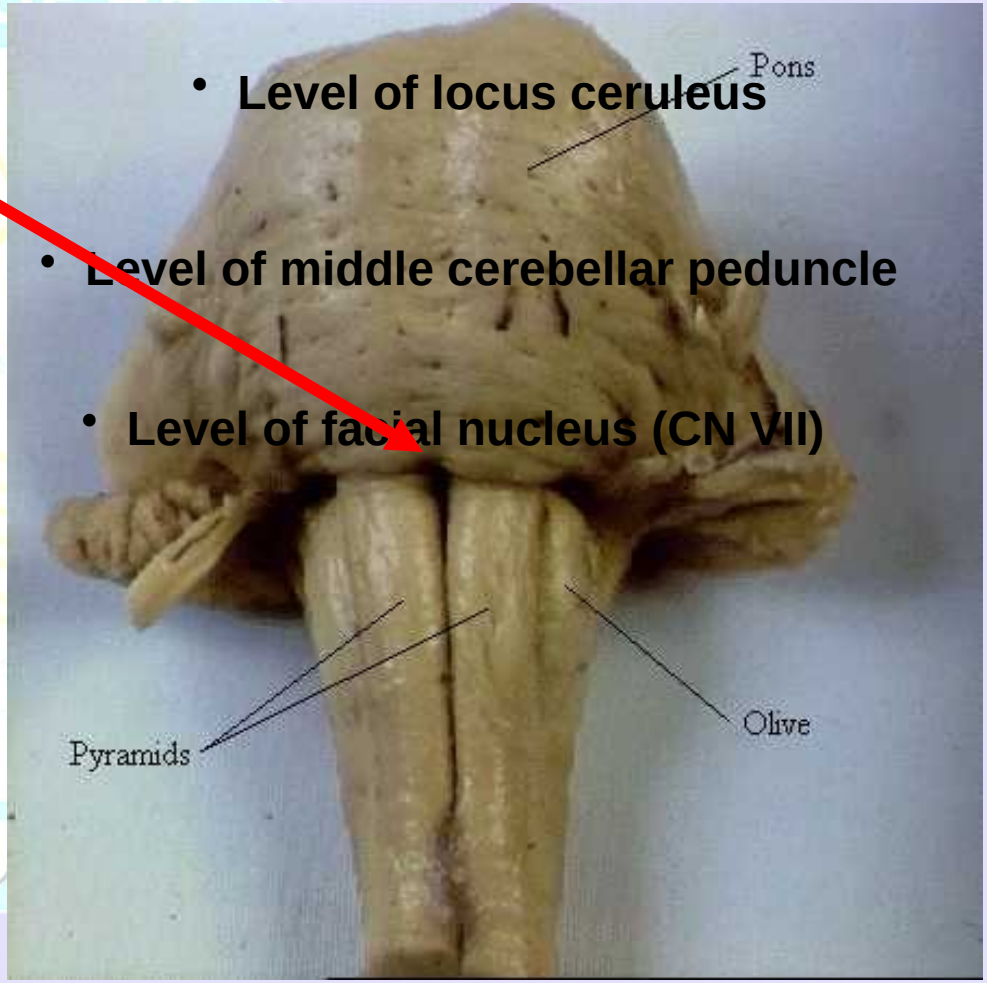
  Contains both longitudinal and
  with pontine nuclei
- Dorsal tegmental portion Co
  formation of the medulla obl



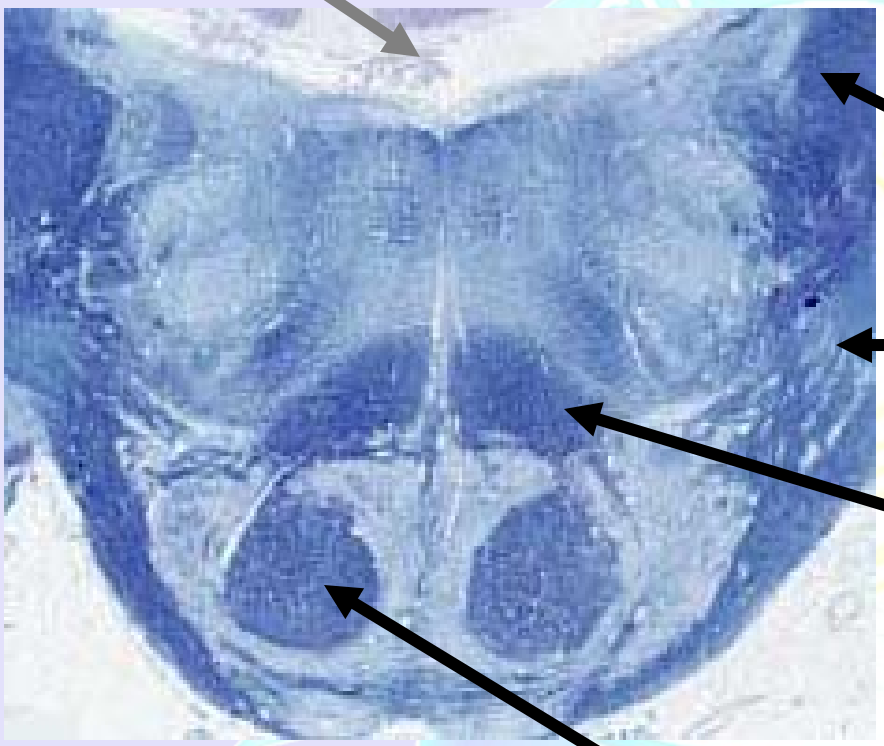
Pontine nuclei
Project to cerebellum
Continuation of arcuate nuclei (over pyramids)

Internal structure of pons

- Structures present at the level of facial nucleus.
- CN VI nucleus – Abducens nerve
- CN VII nucleus – Facial nerve
- CN VIII – Vestibular Nuclei
  - Pure sensory →location lateral



Pons



Connection of pons to cerebellum

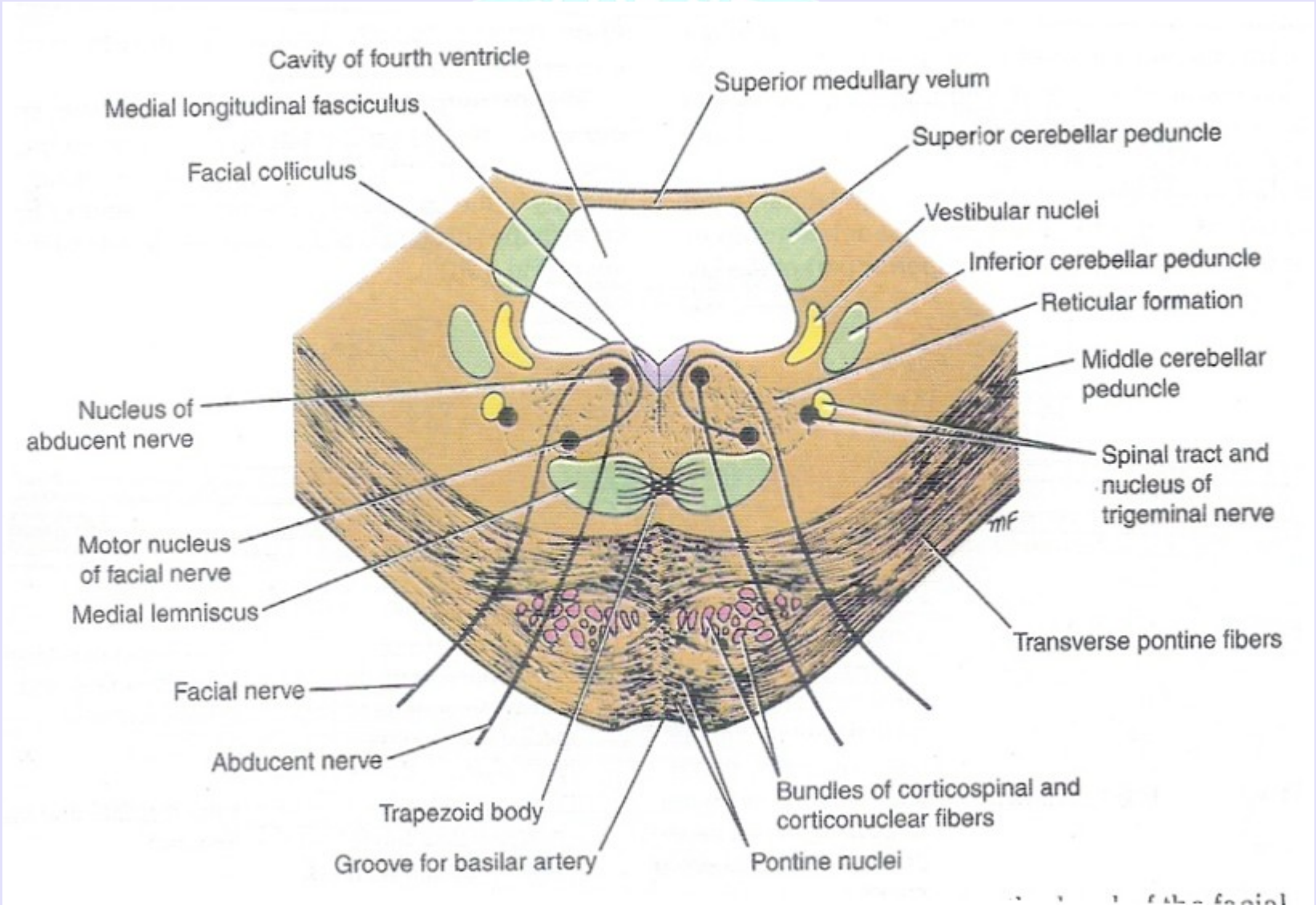
Restiform body (inf. cerebellar peduncle)

Middle cerebellar peduncle

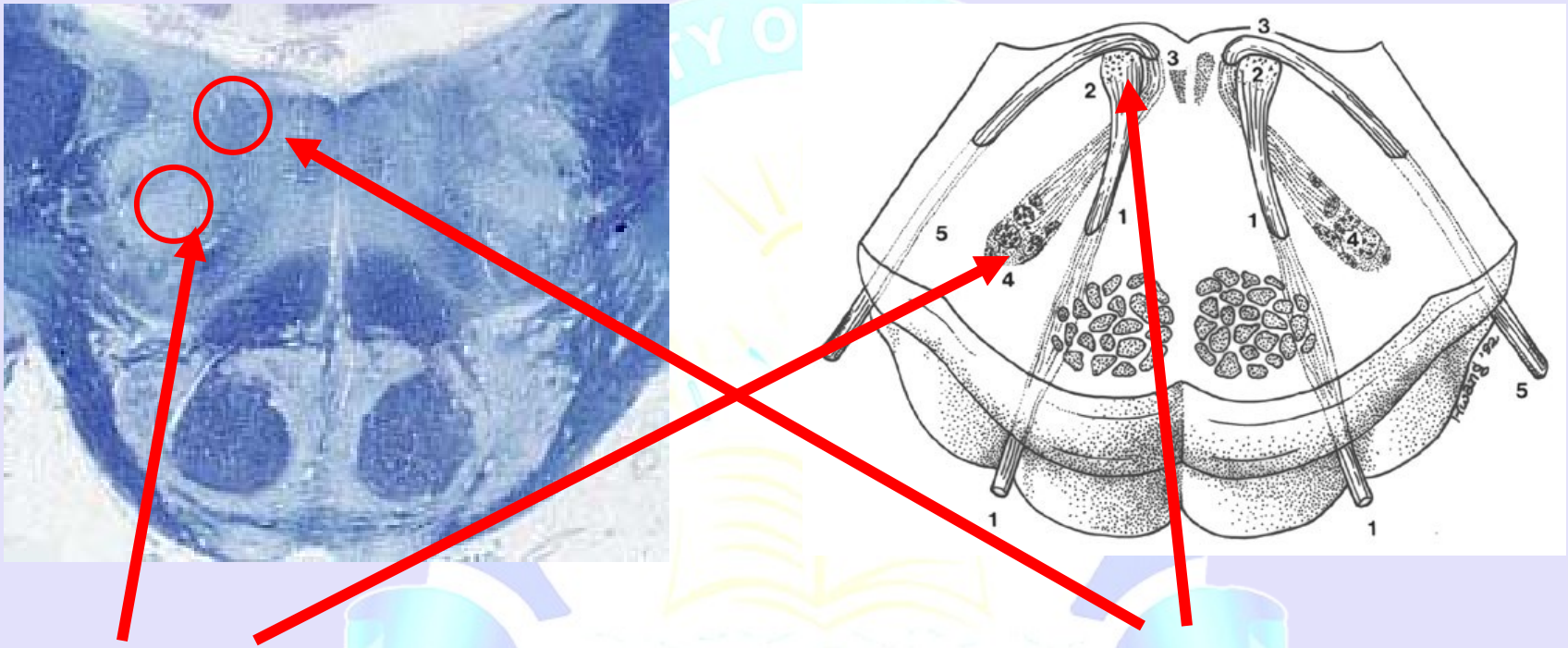
Medial lemniscus
Ascending 2nd order sensory neurons

Descending upper motor neurons

Transverse Section through the caudal part



Abducens nerve and facial nerve



CN VII nucleus – Facial nerve

controls the muscles of facial expression, and functions in the conveyance of taste sensations from the anterior two-thirds of the tongue and oral cavity

CN VI nucleus = Abducens nerve

"somatic efferent" nerve that controls the movement of a single muscle, the lateral rectus muscle of the eye.

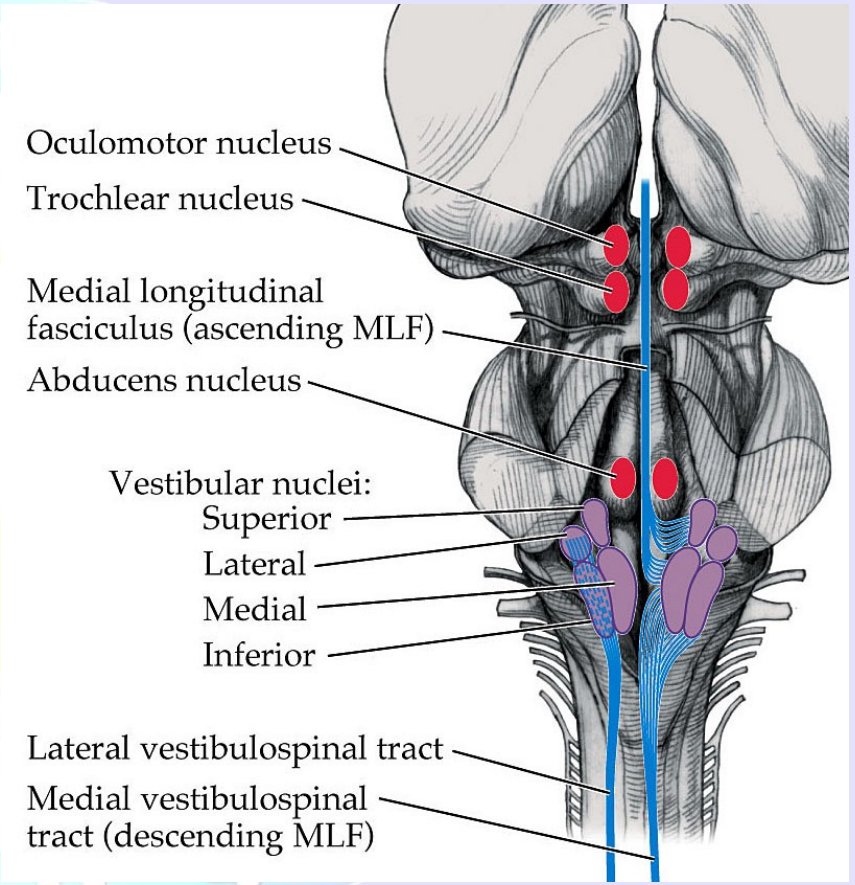
Abduction of eye

Vestibular nuclei



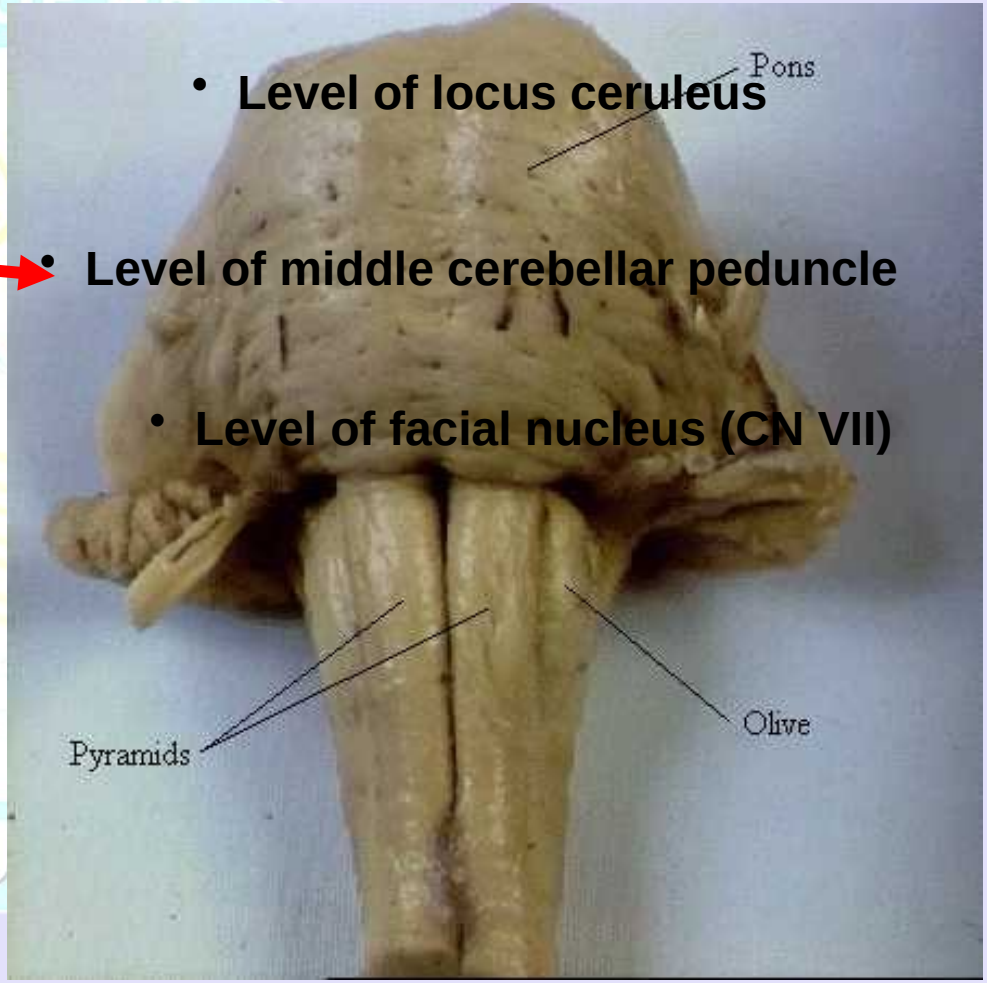
CN VIII – Vestibular Nuclei

Balance

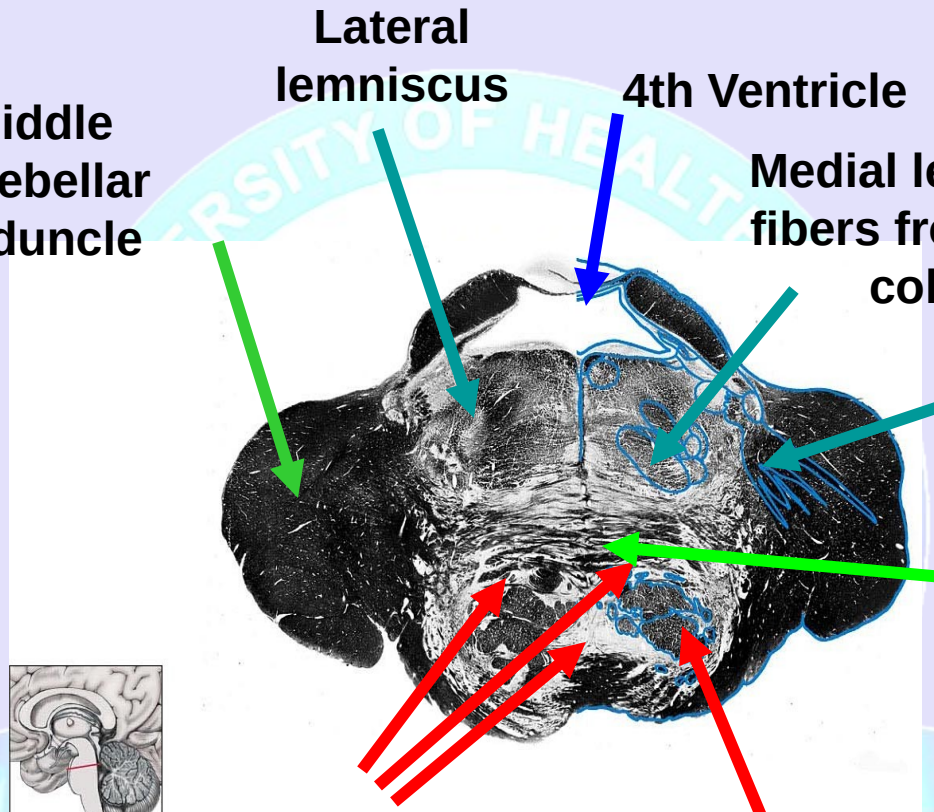


Internal structure of pons

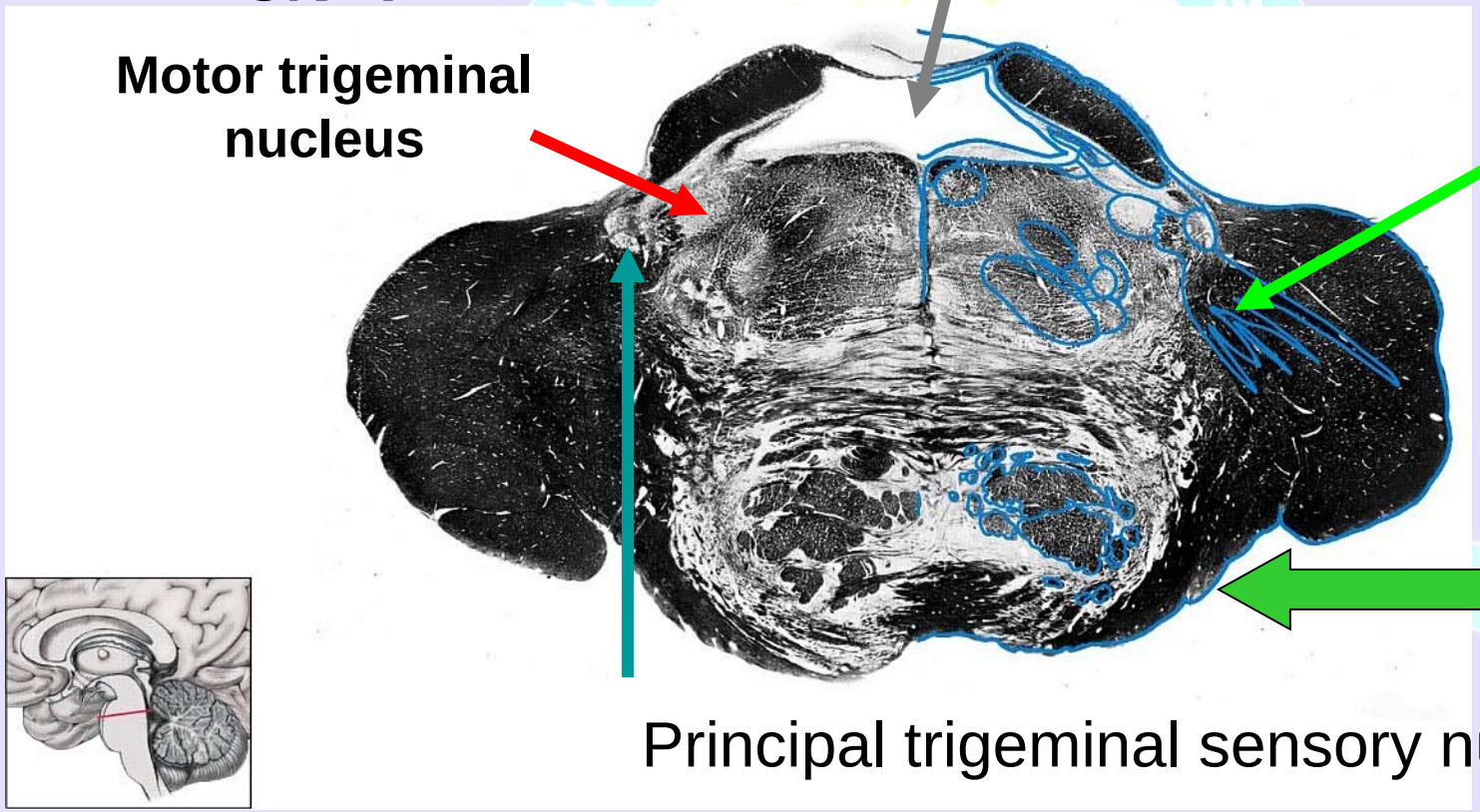
- Structures present at the level of middle cerebellar peduncle.
- Medial lemniscus
- Lateral lemniscus
- Trapezoid body
- Trigeminal nucleus (sensory and motor)
- Pontine nuclei



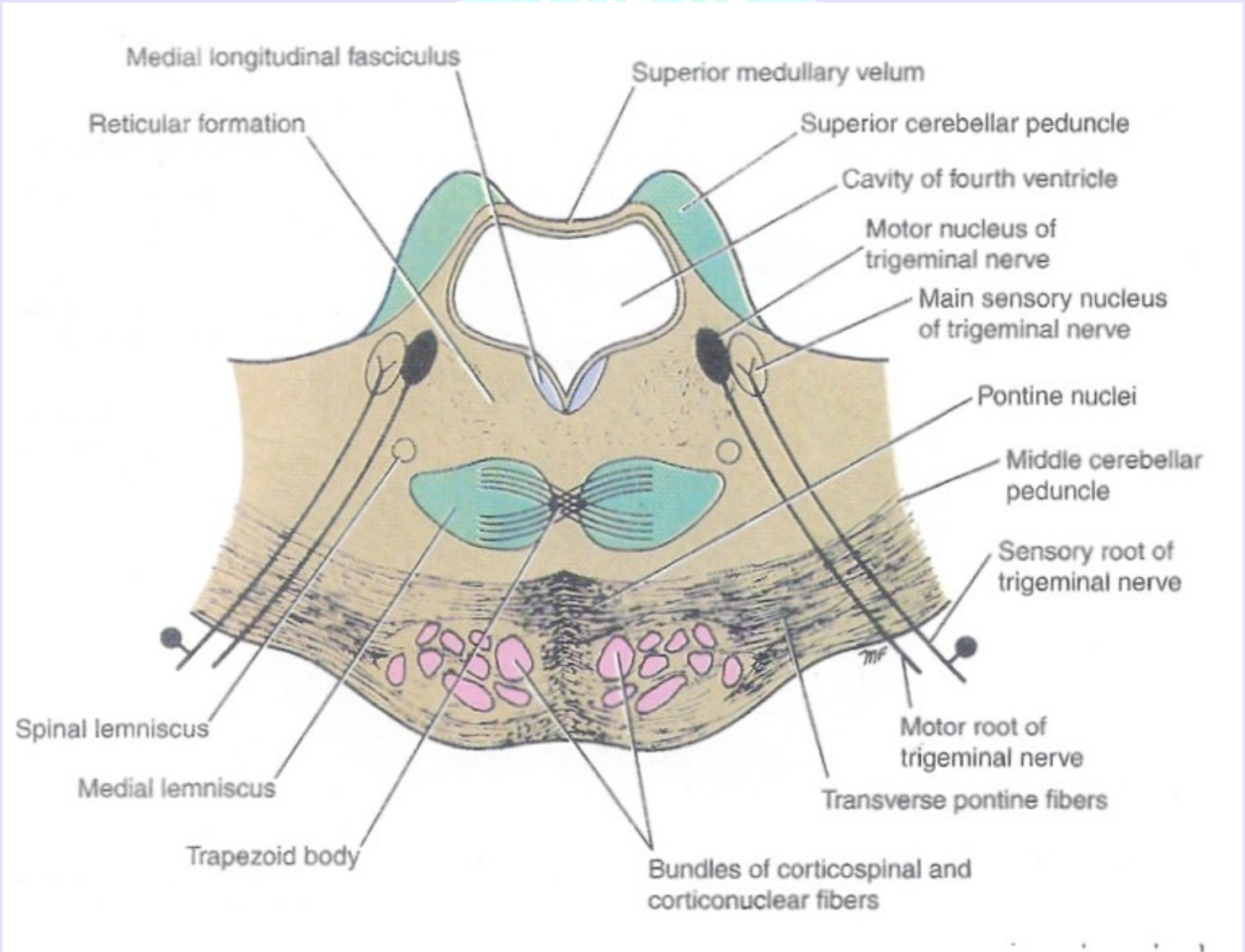
At the level of mid pons



Cranial Nerve of the Mid Pons

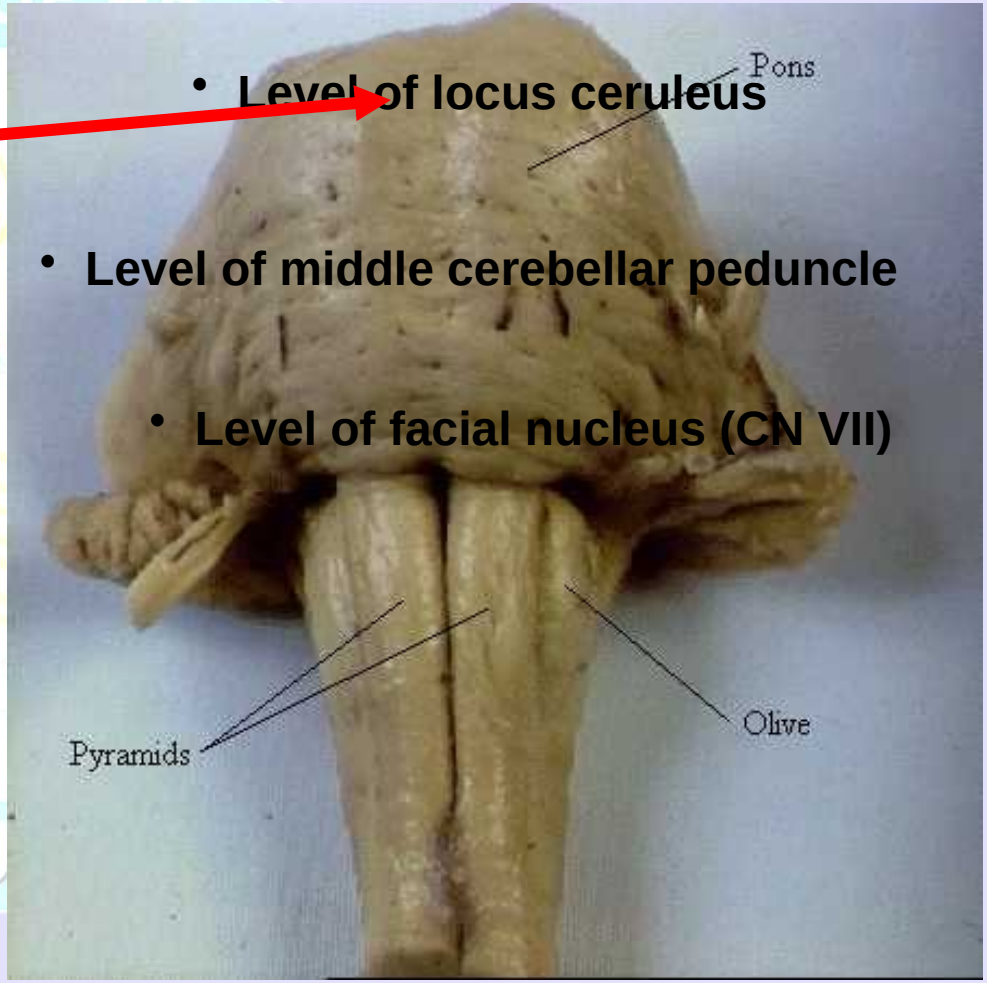


Transverse Section Through The 5th Cranial Nerve:

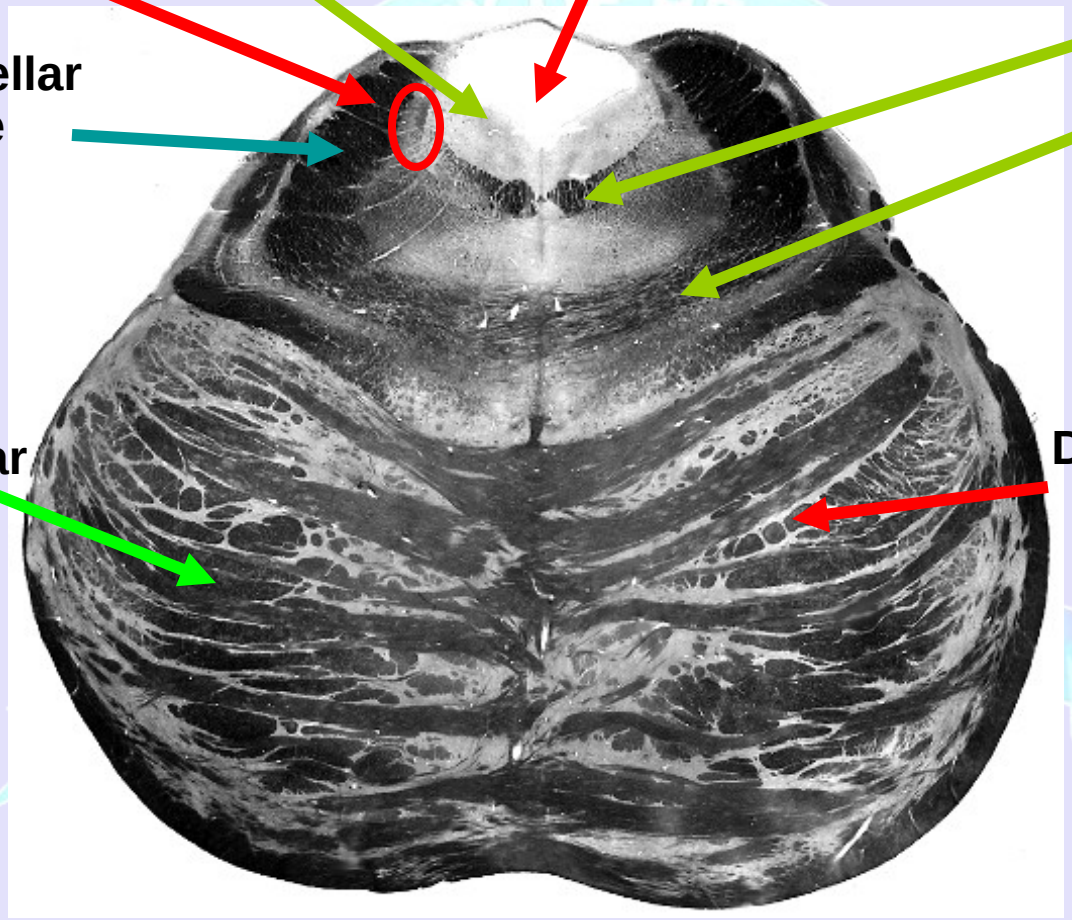


Internal structure of pons

- Structures present at the level of upper pons
- MLF
- Superior cerebellar peduncle
- Locus ceruleus
- Parabrachial nucleus
- Pediculopontine nucleus



Upper pons

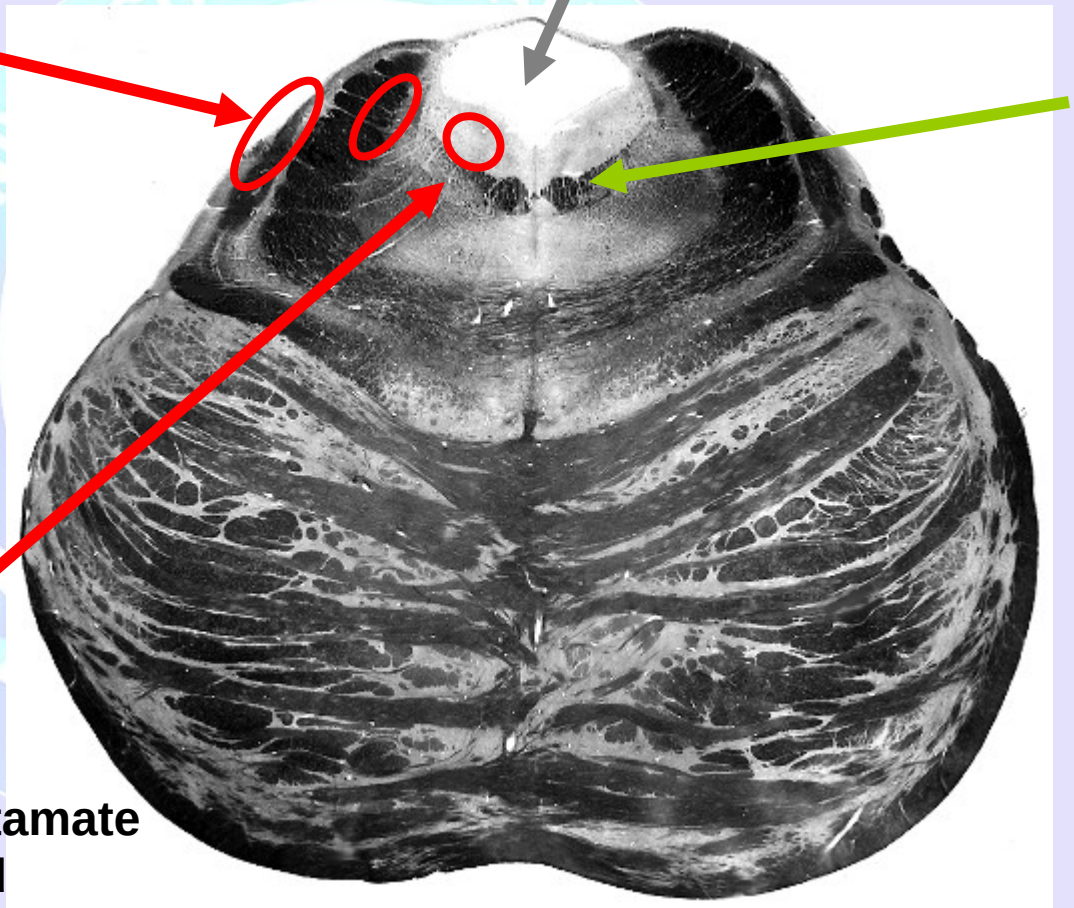

Periaqueductal gray matter
Locus ceruleus
Superior cerebellar peduncle
Transverse ponto-cerebellar fibers
4th ventricle → cerebral aqueduct
MLF
Decussation
Descending upper motor neurons

Upper Pons

4th ventricle → cerebral aqueduct

Parabrachial Nucleus
Also release catecholamines

Neurons also contain melanin



MLF

Pediculopontine Nucleus

Some neurons release acetylcholine
Other neurons release glutamate
They assist in learning and voluntary motor control,e.g. locomotion, saccadic eye

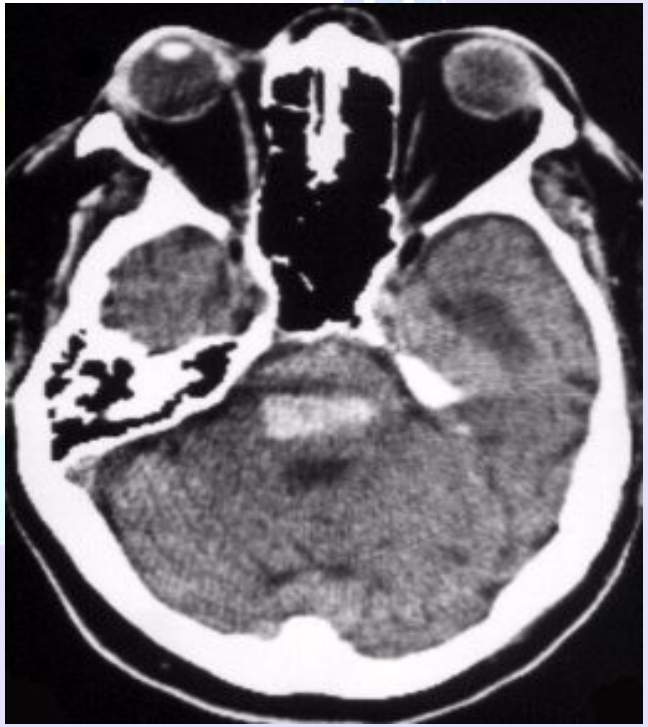
Pontine hemorrhage

- **Pons supplied by basilar artery, anterior, inferior and superior cerebellar arteries.**
- **If hemorrhage is unilateral**
  - **Facial paralysis on the side of lesion (facial nerve nucleus)**
  - **Paralysis of limbs on opposite side (corticospinal fibres)**
  - **Paralysis of conjugate ocular deviation (abducent nerve nucleus)**



Pontine hemorrhage

- **If bilateral**
  - **Pinpoint pupils (ocular sympathetic fibres)**
  - **Bilateral paralysis of face and limbs**
  - **poikilothermia**



Tumors of pons

- Astrocytoma of pons, most common tumor of brainstem

